09 · Does the "€100 -> Gold tier" perk retain people? — RDD (CausalPy)

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> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. Customers who cross **€100** annual spend get **Gold** status and its perks. Gold members retain better — but of course they do, they're big spenders. Does the *perk itself* boost retention, or are we just seeing that big spenders would retain anyway? And where should we set the threshold?

1.0.1 The idea: a sharp rule creates a natural experiment

Regression discontinuity (RDD) exploits the fact that Gold status is assigned by a **sharp rule** on a **running variable** (annual spend): below €100 you're out, at €100-or-above you're in. Now compare a customer at €99.50 with one at €100.50. On *retention potential* they are essentially identical — nobody is meaningfully more loyal for having spent one euro more. The *only* thing that differs between them is that one got the perk and one didn't. So **any jump in retention exactly at €100** must be caused by the perk, not by the underlying "big spenders retain more" trend (which is smooth through the cutoff). RDD reads off the size of that jump.

1.0.2 What it can and can't tell you

- It estimates a **LATE at the cutoff** — the perk's effect *for customers near €100* (here *local* means local-to-the-cutoff: an ATE exactly at €100, **not** the complier-LATE of a fuzzy design). It says nothing about your €500 whales; don't extrapolate the jump to them.
- It's only valid if customers can't **manipulate** their spend to just clear €100 (that would make the people just above and just below systematically different). We test this with a **simplified McCrary-style** density check (a symmetric-window binomial test for a suspicious pile-up of customers just above the cutoff; the full McCrary 2008 test fits a local-linear density with a Wald SE).
- A headline RD number is only as good as its **robustness checks**: does the jump survive different **bandwidths** (how wide a window around €100 we use), **placebo cutoffs** (fake thresholds should show no jump), and **polynomial order** (linear vs curved fits)? We run all of them.

> **On real data.** RDD fits anywhere a **threshold rule** assigns a treatment: loyalty tiers,

credit-score cutoffs for an offer, "spend €X for free shipping," scholarship/eligibility scores. You need the running variable, the cutoff, and the outcome — all of which live in your CRM.

It follows the repo's fixed **7-step contract**: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

1.1 2 · Simulate a ground truth

Retention rises smoothly with annual spend (big spenders retain more anyway) — **plus** a true **+14pp jump** exactly at €100 from the Gold perk. The smooth part is the confounder; the jump is the causal effect. Spend is concentrated around the cutoff so both sides are well-populated.

```
TRUE perk effect = +14% retention at €100  ·  50% are Gold
```

	spend	treated	retention
0	84.325206	0	1.0
1	101.996816	1	0.0
2	93.392393	0	1.0
3	93.281224	0	0.0
4	102.504635	1	0.0

1.2 3 · Identify — a jump at the cutoff is causal (and its validity conditions)

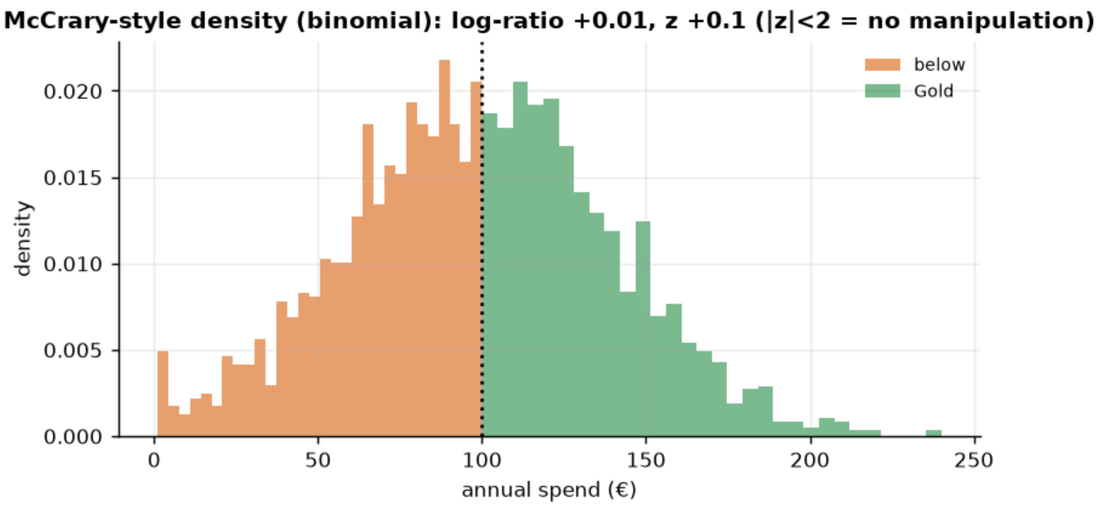
Sharp RD estimand: $\tau_{RD} = \lim_{x \downarrow c} \mathbb{E}[Y \mid X{=}x] - \lim_{x \uparrow c} \mathbb{E}[Y \mid X{=}x]$ — the discontinuity in retention at spend = €100. **Identification:** potential-outcome means are *continuous* at the cutoff, so units just either side are comparable and the jump is causal. This is a **LATE at the cutoff** — local to €100 (*local* = local-to-the-cutoff, i.e. an ATE exactly at €100, **not** the complier-LATE of a fuzzy design). Two things must hold, both checked below:

- **No manipulation** of the running variable at the cutoff (customers/business can't sort across it) — a simplified McCrary-style density check (a symmetric-window binomial test).
- **The functional form isn't driving the jump** — checked via bandwidth, placebo cutoffs, and polynomial order.

1.3 3b · Validity check 1 — McCrary-style density check (manipulation)

If customers game their spend to just clear €100 (or the business nudges them), the *density* of spend jumps at the cutoff and RD breaks. We compare the density just below vs just above with a symmetric-window binomial test. (This is a *simplified* version; the full McCrary 2008 test fits a local-linear density on each side with a Wald SE.)

```
density just below 23.7 vs above 24.0, log-ratio +0.01, binomial z +0.15 - no
significant sorting across the cutoff (|z| < 2).
```



(The density is smooth through €100 — no suspicious pile-up of customers *just* above the cutoff — so we have no evidence of manipulation, and the RDD comparison is fair.)

1.4 4 · Estimate — Bayesian regression discontinuity

Now the estimate itself. We fit a **local linear** regression on each side of the cutoff — a straight line for the below-€100 customers and another for the Gold customers — and the **height of the step where they meet at €100** is the causal effect of the perk. We restrict to a **bandwidth** of +/-€40 around the cutoff: the customers near the threshold are the comparable ones (a €98 and a €102 customer are alike; a €500 whale is not). Wider bandwidths borrow non-comparable customers; narrower ones throw away data — Step 5 checks the answer is stable to that choice.

We model retention with a local linear probability model — standard for RD jumps; a logit link changes nothing material because the fit is local.

```
RD jump +11.3% (true +14%)  ·  90% credible interval [+3.8%, +18.7%]
RDD convergence: max r-hat 1.000 - min ESS 1378 - divergences 0
```

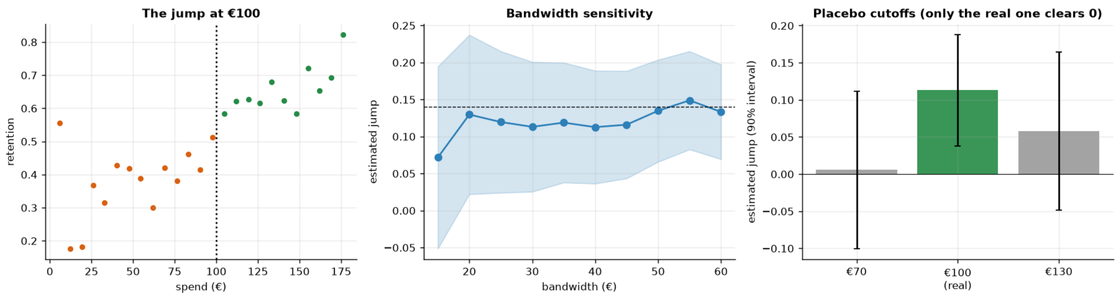
1.5 5 · Validate — see the jump, plus bandwidth / placebo / polynomial robustness

A credible RD number is one that **survives** the researcher-degrees-of-freedom checks:

- **Bandwidth curve** — the estimate should be stable across a sensible range of bandwidths.
- **Placebo cutoffs** — run RD at fake thresholds (€70, €130) where there's no perk; the jump should be ~ 0. A real jump at a fake cutoff would mean the method finds discontinuities in noise.
- **Polynomial order** — linear vs quadratic local fit shouldn't move the answer much.

```
Bandwidth sweep: +7.2% … +14.9% across bw 15-60 (mean +12.0%) - drifts mildly
but stays clearly positive.
Polynomial: linear +11.3%, quadratic +10.9%.
Placebo cutoffs (jump, t-stat): €70 +0.6% (t=+0.09), €130 +5.8% (t=+0.90)
```

```
Largest placebo |t| = 0.90 - all statistically indistinguishable from zero; the
real €100 jump's 90% CrI [+3.8%, +18.7%] excludes zero.
```



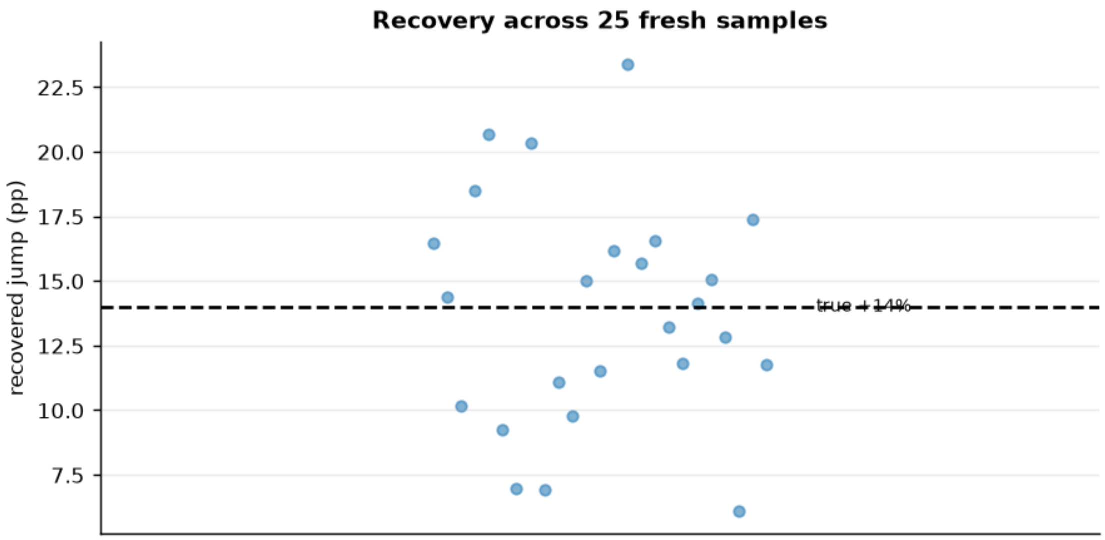
How to read the robustness battery. *Left* — bin retention by spend and you can *see* the step at €100 sitting on top of the smooth "big spenders retain more" trend; that step is the perk effect. *Middle* — the estimate **drifts mildly** with the bandwidth (narrow windows are noisier, wider ones pull in curvature) but stays clearly positive and brackets the truth across the whole range, so we're not cherry-picking a window. *Right* — the decisive check, now **with 90% intervals**: re-run the same local-linear RD at **fake** cutoffs (€70, €130) where no perk exists. Their point estimates wobble around zero — the €130 placebo is the largest, but read its *interval*, not its bar height: it comfortably spans zero (small-sample wobble in a thinner slice of data), and every placebo's |t| is well under 2. Only the **real** €100 cutoff has an interval that **excludes** zero. That is the honest test — a *t*-stat, not the eyeball. Together with the McCrary density and the linear-vs-quadratic agreement, it's the full case that the jump is a real causal effect, not an artefact of how we drew the lines.

(Implementation note: for speed, the battery's refits are local-linear ***OLS approximations*** *of the same specification, judged on frequentist t-stats; the headline €100-cutoff estimate itself stays fully Bayesian.)*

1.5.1 5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test is whether the estimator is **centred on the truth over repeated samples** and whether its 90% interval **covers** the truth at the stated rate. We refit on many fresh samples (a small fast fit each) and check both.

```
RD jump across 25 seeds: mean +13.8% (true +14%)  bias -0.2%  sd 4.3%  ·  90% CI
covers truth in 23/25 seeds - near-nominal calibration.
```



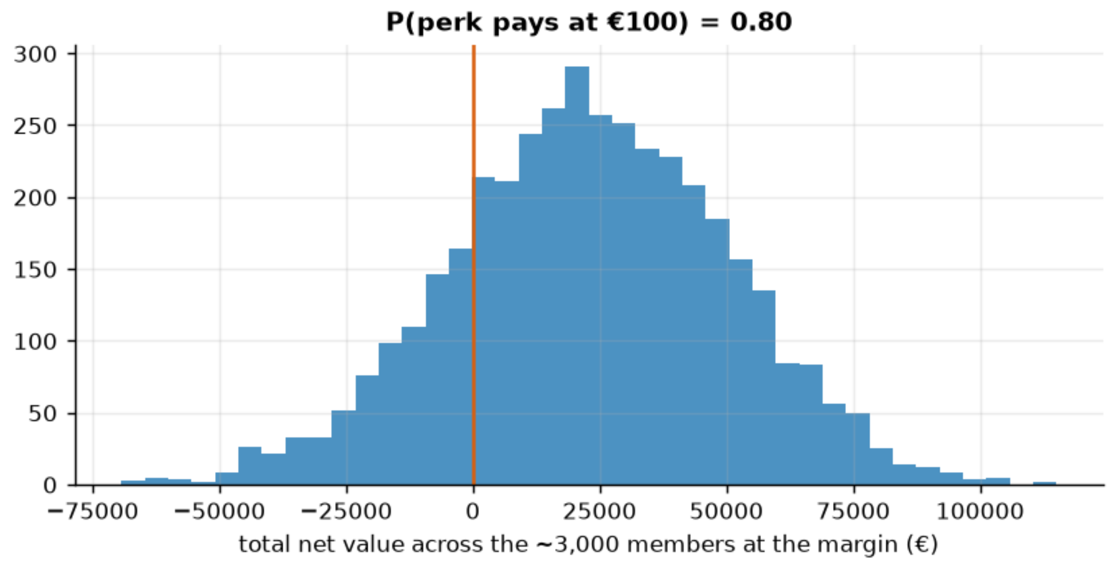
1.6 6 · Decide, in euros — does the perk pay, and where to set the cutoff?

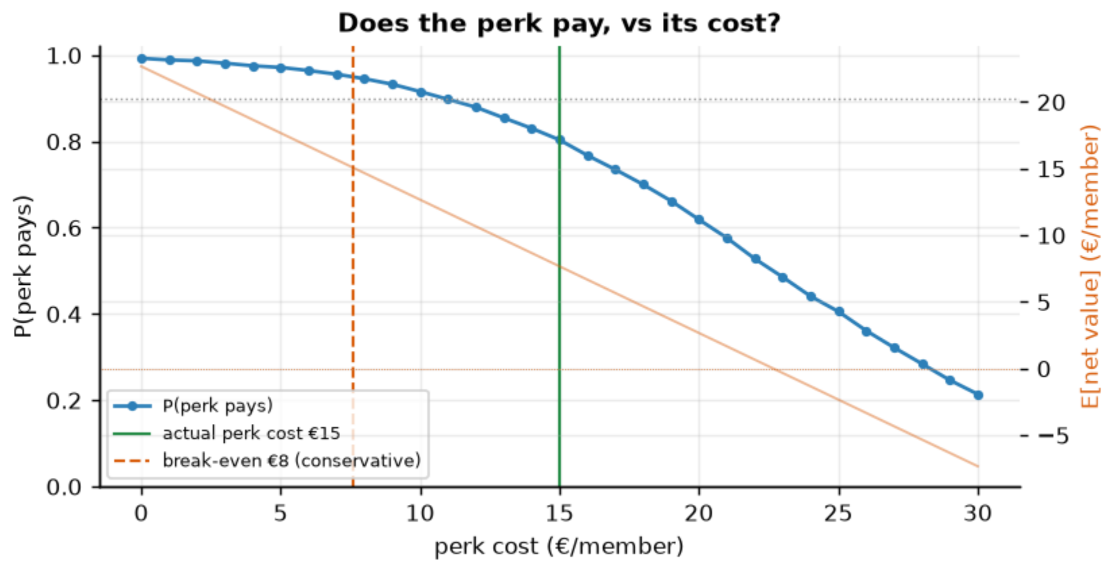
Value the retention lift: extra retained customers x annual value, minus the perk cost, **at the current €100 margin** (that's what RD identifies). We also sweep the perk cost to find the break-even.

> **What RD can and can't answer.** RD identifies *only* the local effect at the €100 margin — it tells us whether the perk pays **there**, but it **cannot** tell us where to set the cutoff (moving the threshold extrapolates beyond the RD's local validity; that question needs an experiment or multi-cutoff data).

> **Two stated inputs.** `N_AT_MARGIN ~ 3,000` is an illustrative count of members inside the +/-€40 window around €100 per cycle — swap in your CRM's number; the **0.9** go/no-go bar is the cookbook's convention for "sure enough to act without a further test".

```
Net €7.7/member  ·  total €22,960 [90% €-22,140, €67,444]
P(perk pays) 0.80 -> reconsider  ·  break-even perk cost ~ €8/member
(conservative).
Perk-cost sweep €0-30: P(perk pays) falls 0.99 -> 0.21; at the actual €15 cost
P(pays) = 0.80, E[net] = €8/member (break-even €8).
```





How to read the decision histogram. Mass to the right of the orange zero line = P(perk pays); the wide spread — *not* the positive mean — is what makes this a *reconsider / TEST* call.

1.7 7 · Caveats

- **RD is *local.*** The effect is identified only *at the cutoff* — the perk's effect on customers near €100, not on your €500 whales. Don't extrapolate the jump to everyone.
- **Manipulation breaks it.** We checked the McCrary density; watch for heaping at round numbers.

- **Bandwidth is bias–variance.** The estimate drifts mildly with the window (narrow = noisier, wide = more curvature) but stays clearly positive here; an estimate that *swings across zero* with bandwidth would be a red flag.
- **Fuzzy RD if the rule isn't sharp.** If crossing €100 only *raises the probability* of Gold (grandfathering, overrides), use the fuzzy-RD ratio (jump in Y ÷ jump in P(treated)) — this ratio is exactly the Wald/IV estimator of notebook 11, with the threshold acting as the instrument.